Experiment 4

Simulating a Network with Different Devices

Lab 4 Objectives:

● Simulate a network with multiple devices using network simulation software

Prerequisites: The prerequisites of this lab are

1. Basic understanding of networking concepts (IP addressing, subnets, routing).
2. Familiarity with network simulation tools such as Cisco Packet Tracer, GNS3, or NetSim.
3. Knowledge of networking protocols (TCP/IP, ICMP, HTTP, etc.)

Outcomes: By the end of this lab, students will be able to:

1. Design and configure a network topology with different types of devices.
2. Assign and verify IP configurations in a simulated network environment.
3. Monitor and analyze network traffic to understand data flow.

Part 1: Setting Up the Network Simulation

Step 1: Selecting Network Devices

Open Cisco Packet Tracer.

Drag and drop the following devices:

- 1 Router (Router-PT)
- 1 Switch (2960 Switch)
- 2 PCs (PC0, PC1)
- 1 Server

Step 2: Establishing Network Connections

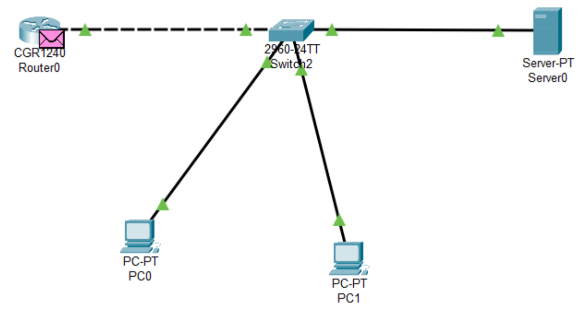
Use Copper Straight-Through cables to connect:

✓ PC0 to Switch (FastEthernet0/2)
✓ PC1 to Switch (FastEthernet0/3)

- ✓ Server to Switch (FastEthernet0/4)

Use Copper Straight-Through cable to connect Switch to Router:

Switch FastEthernet0/1 to Router GigabitEthernet0/0



Part 2: Configuring IP Addresses

Step 1: Assigning IPs to End Devices

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.11	255.255.255.0
Server	192.168.1.20	255.255.255.0

Step 2: Configuring Router Interfaces

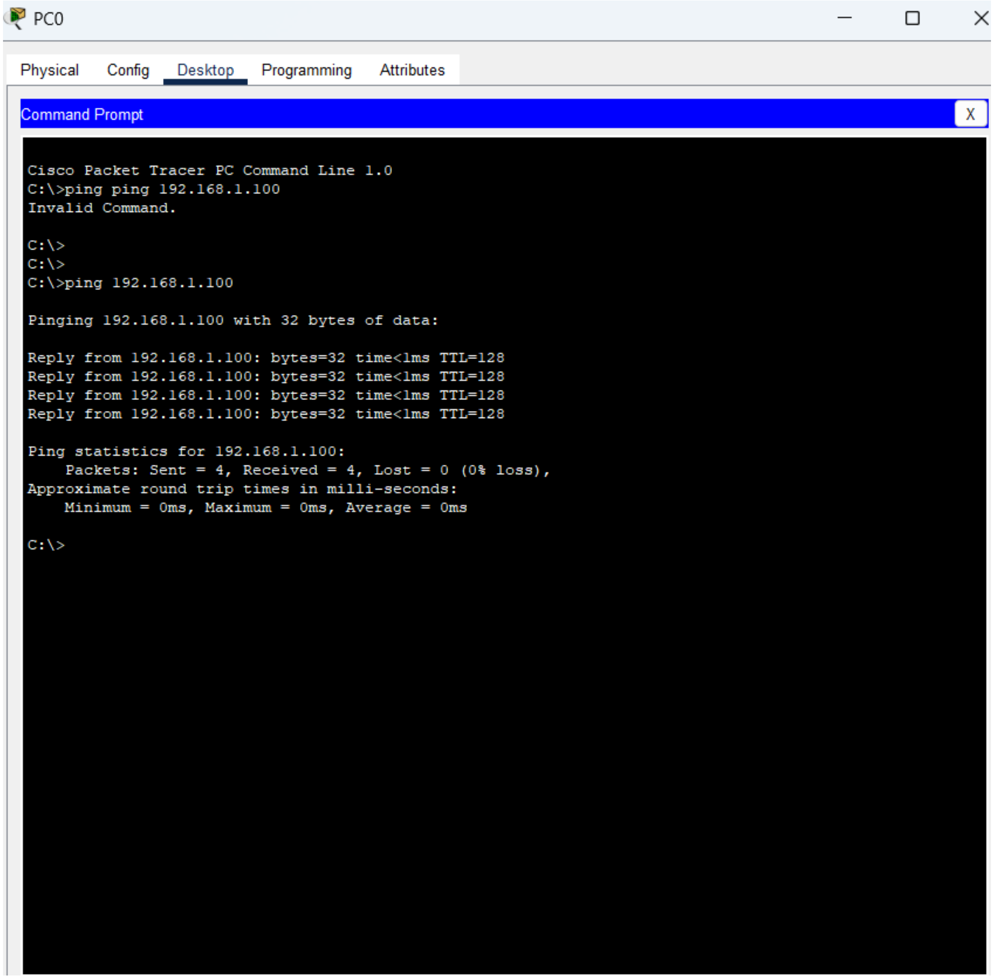
- Access the router CLI and enter the following commands:

Router> enable
Router# configure terminal
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router# write memory

Part 3: Verifying Network Configuration

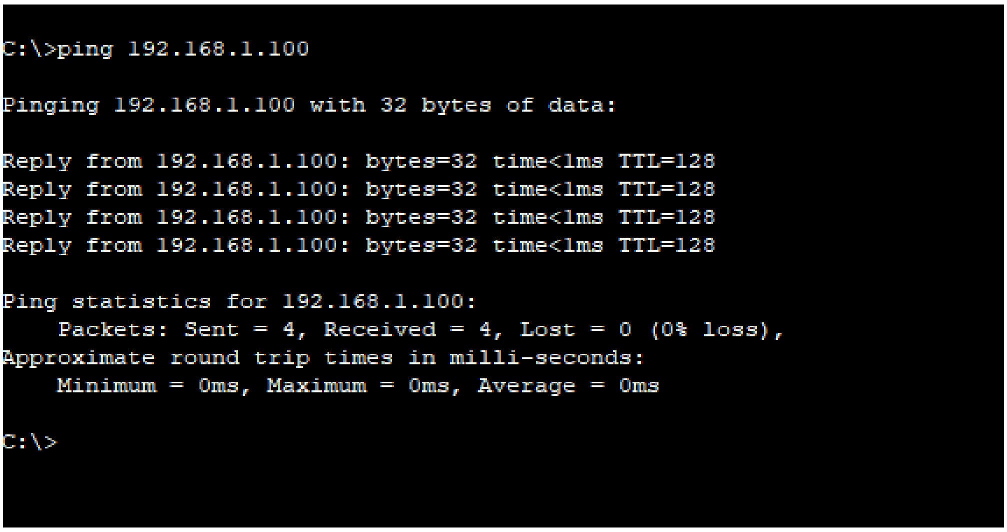
Step 1: Testing Network Connectivity

- From PC0, open the command prompt and ping PC1:




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping ping 192.168.1.100
Invalid Command.

C:\>
C:\>
C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

From PC1, ping the Server:



```
C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
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Ping statistics for 192.168.1.100:
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Approximate round trip times in milli-seconds:
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C:\>
```

Part 4: Analyzing Data Flow

Step 1: Capturing Network Traffic

To capture network traffic, we use tools like **Wireshark** or the **simulation mode of Cisco Packet Tracer**.

- **Wireshark** is a widely used packet analyzer that captures and analyzes real network traffic.
- **Cisco Packet Tracer's simulation mode** allows us to view packet flow within a virtual network.

Process of capturing packets:

1. Initiate data transfer between network devices (for example, by using the ping command).
2. Packets are captured by Wireshark or the Packet Tracer simulation tool.
3. By applying filters, we can view only specific protocol packets, such as ICMP.

Filtering ICMP packets:
ICMP is the protocol used for ping requests and replies. By typing icmp in the filter bar, only ping-related packets will be displayed.

This process helps us analyze network traffic to understand data flow and verify network connectivity.

Step 2: Tracing Data Flow

- From PC0 command prompt, use traceroute to PC1:

```
C:\>tracert 192.168.1.100

Tracing route to 192.168.1.100 over a maximum of 30 hops:

  1   4 ms      4 ms      4 ms      192.168.1.100

Trace complete.

C:\>
```

Top